ROMA TRE
UNIVERSITÀ DEGLI STUDI

iNGEGNERIA
DIPARTIMENTO DI ECCELLENZA

Information Visualization

Orthogonal Drawings from Visibility Representations

M. Patrignani

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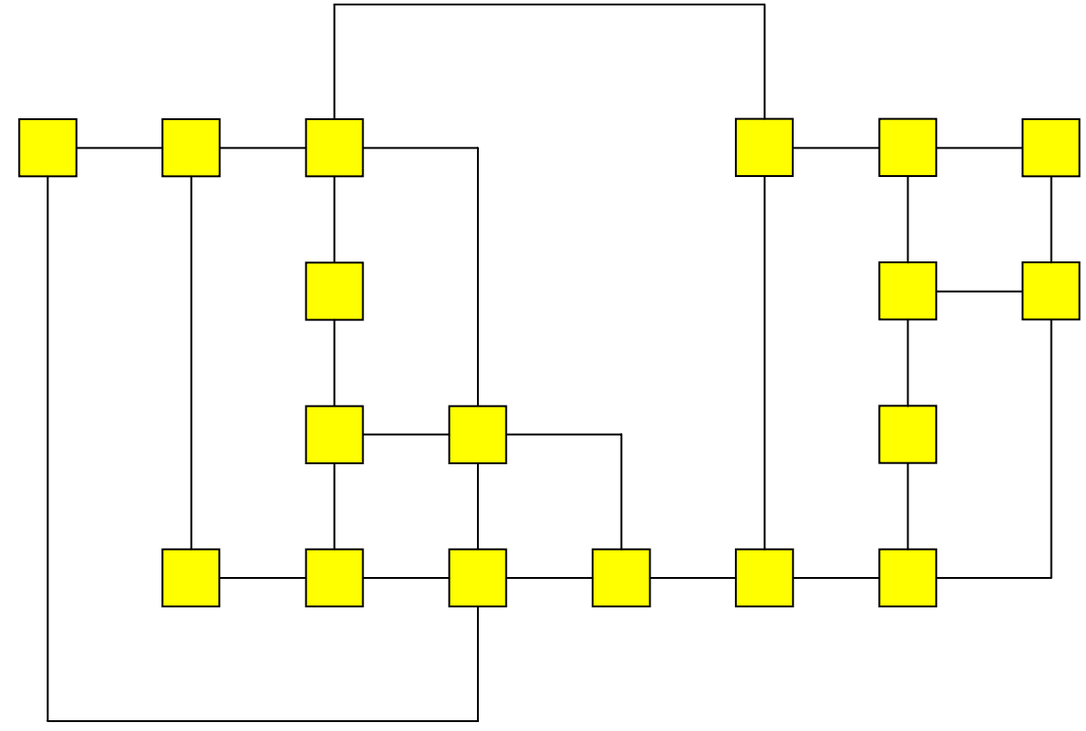
Orthogonal drawings

- Orthogonal drawings obtained with network flow algorithms
  - tend to have large empty faces
  - tend to be self-similar
    - some experiments show that their fractal dimension is about 1.7 [Patrignani 04]
  - tend to push all bends on the same edges

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Orthogonal drawings by network flow

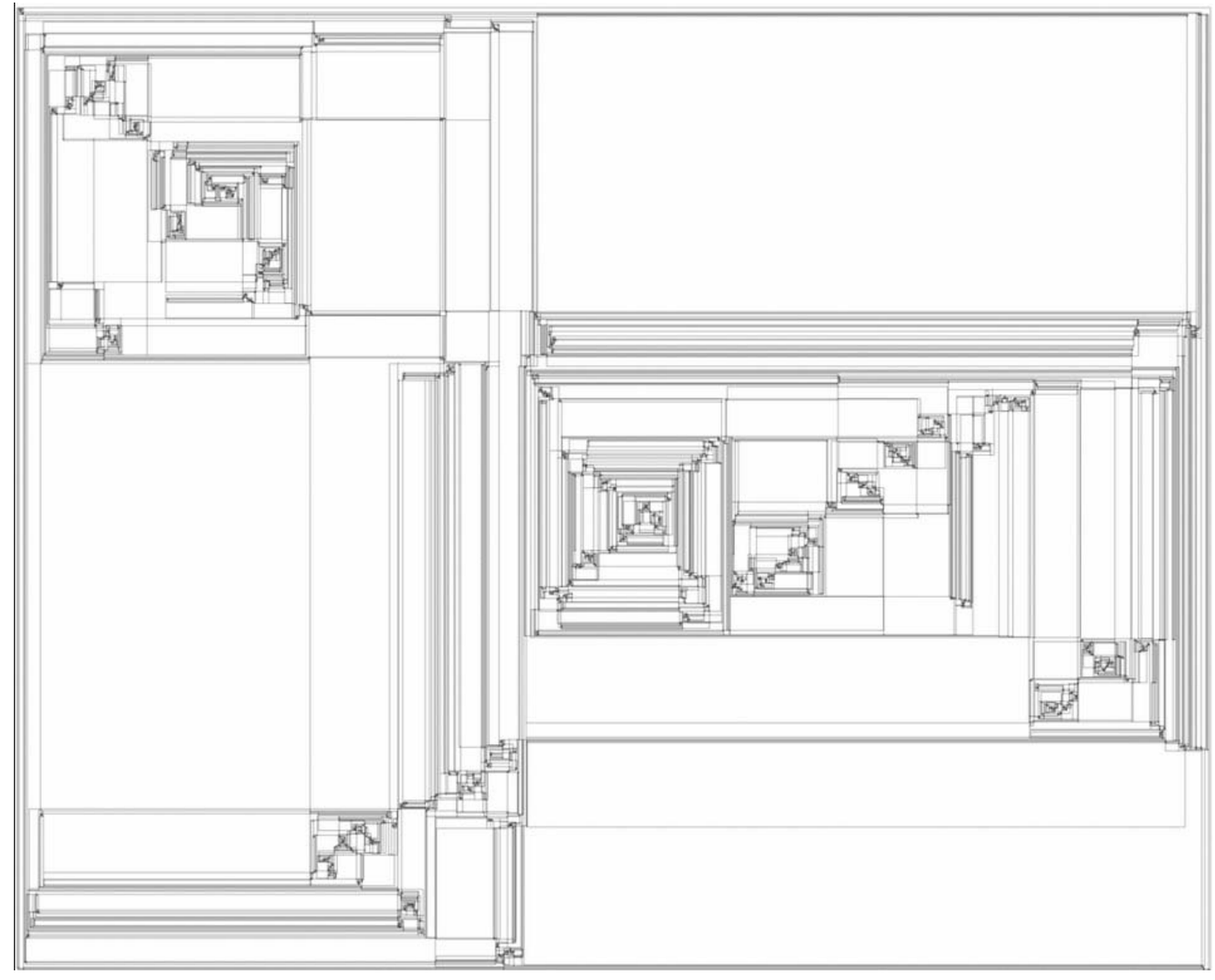
- An orthogonal drawing of a small graph obtained with a network flow technique



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Orthogonal drawings and empty faces

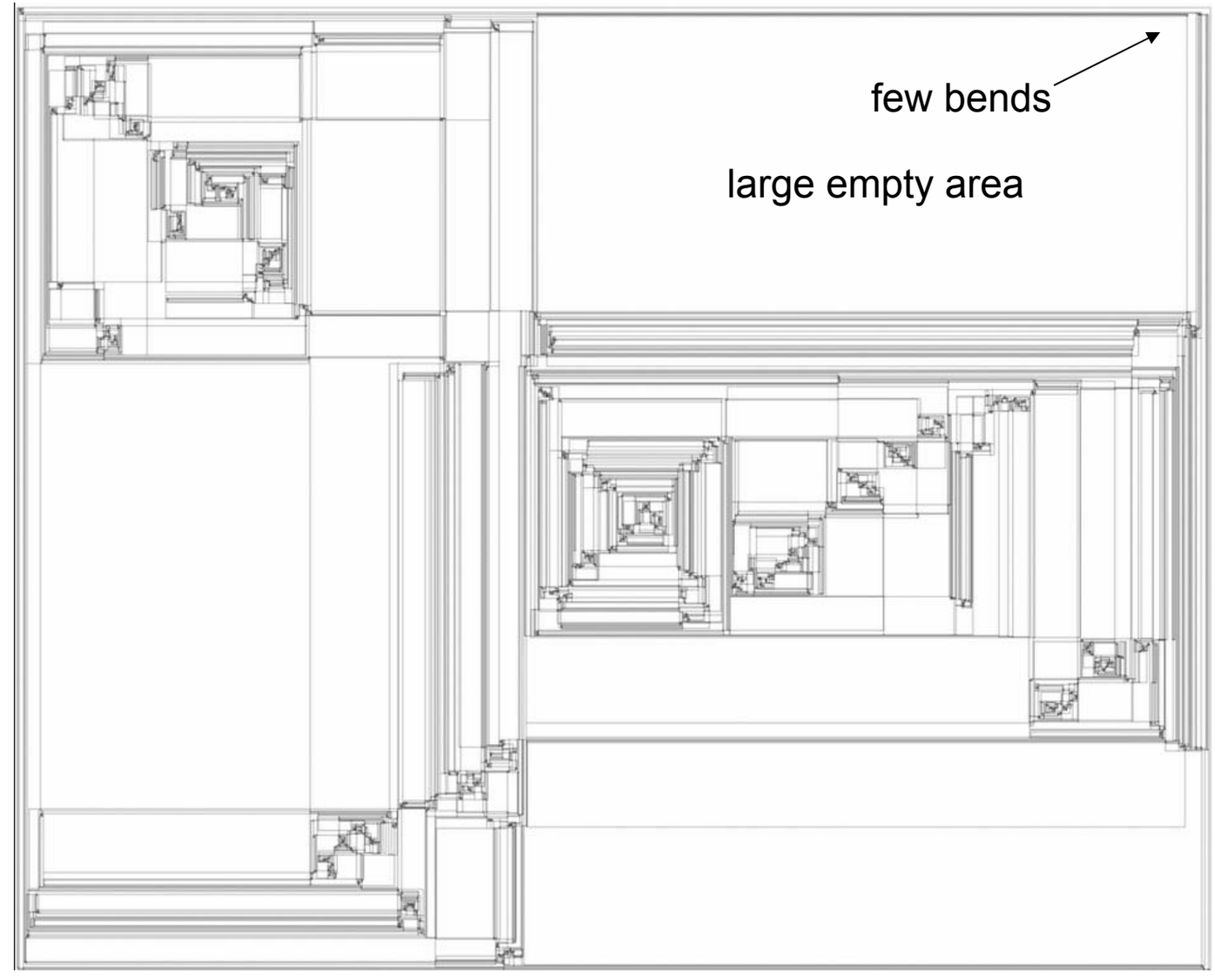
- A random simple, planar, triconnected graph with 5,046 nodes and 10,095 edges
  - generated by the Pigale library
  - drawn with Tamassia's algorithm
    - 3,844 bends



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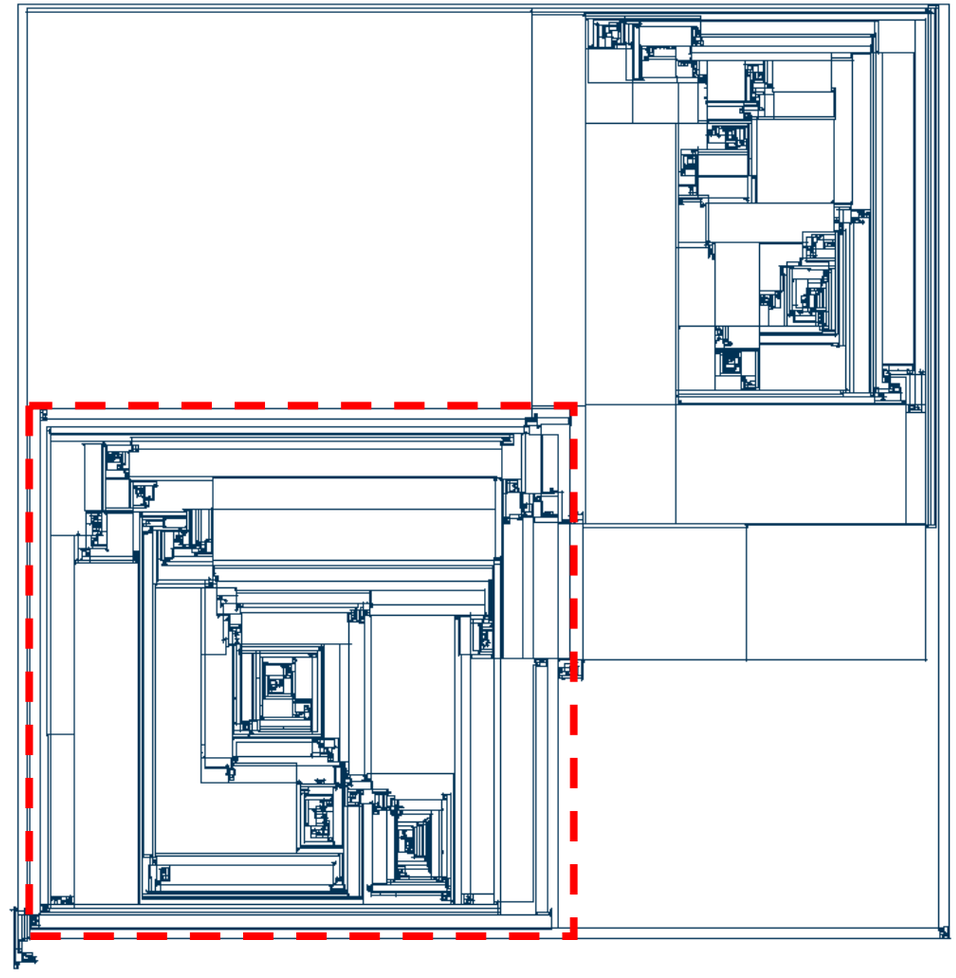
Orthogonal drawings and empty faces

- An edge that has few bends bounds a region of the plane that is unlikely to fill with some other portion of the graph



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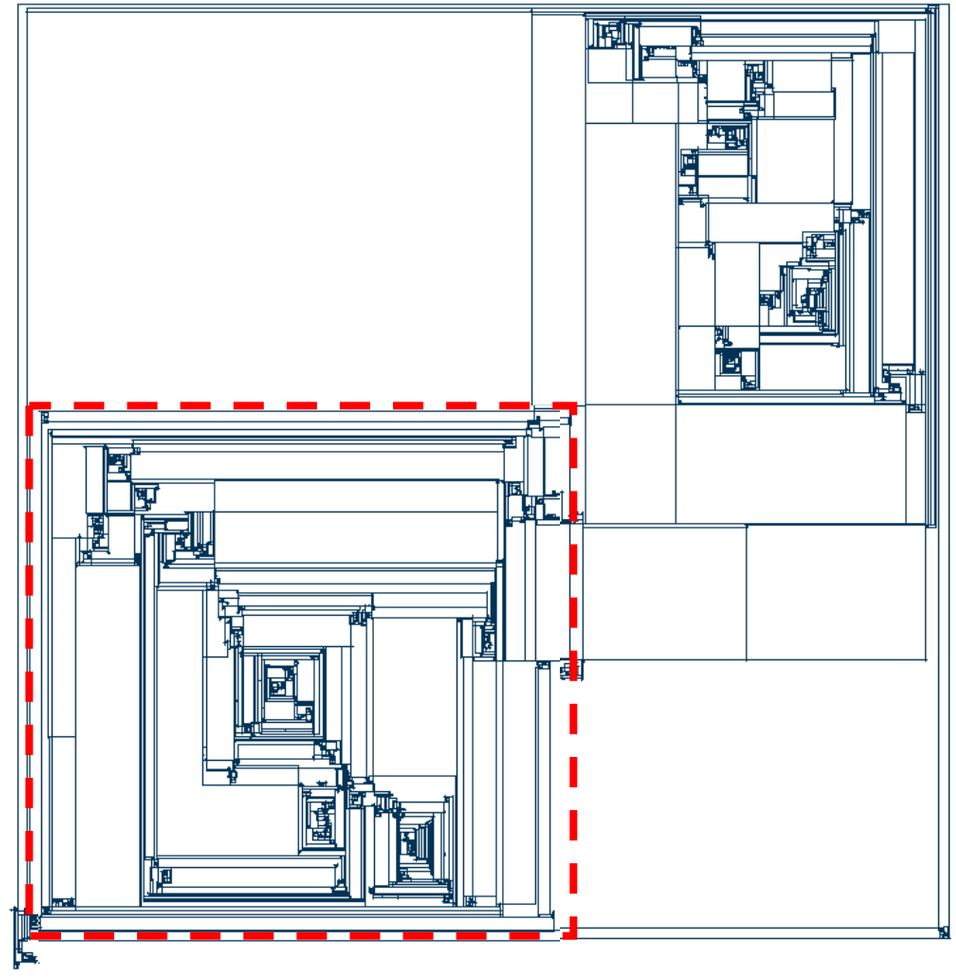
Orthogonal drawings and self similarity



- A random simple, planar, connected graph with 4,700 nodes and 6,155 edges
  - generated by the Pigale library
  - drawn with Tamassia's algorithm
    - 2,463 bends

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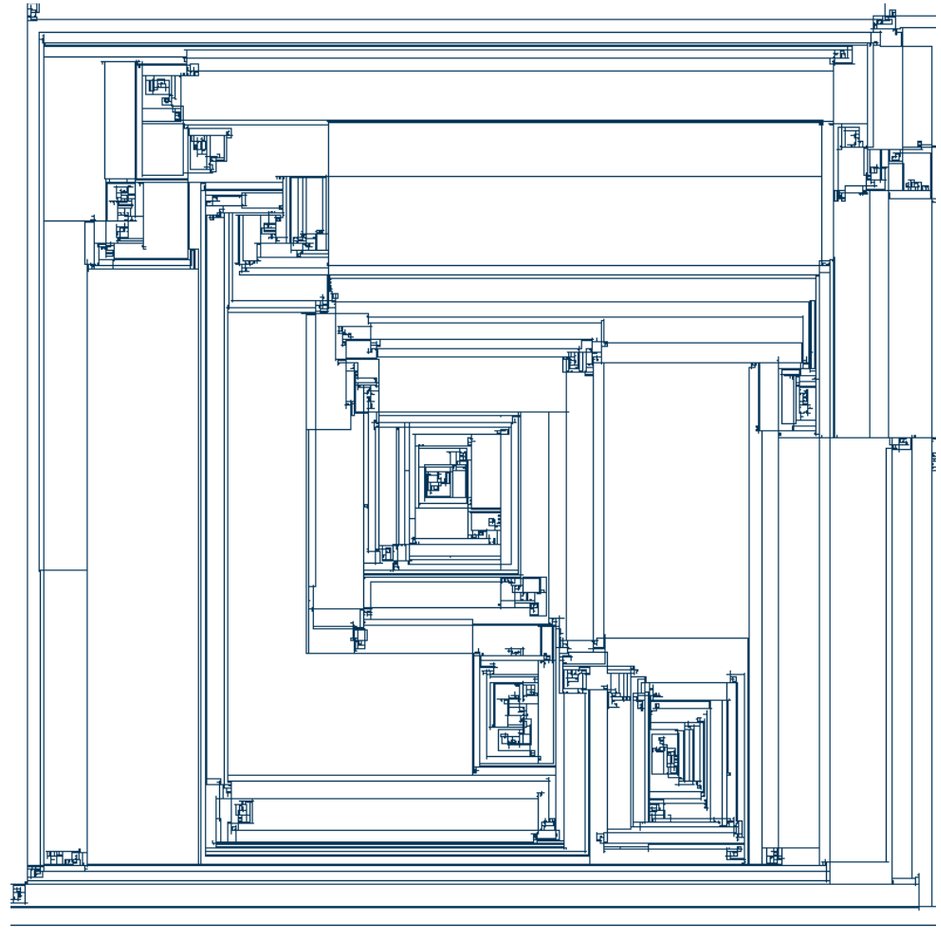
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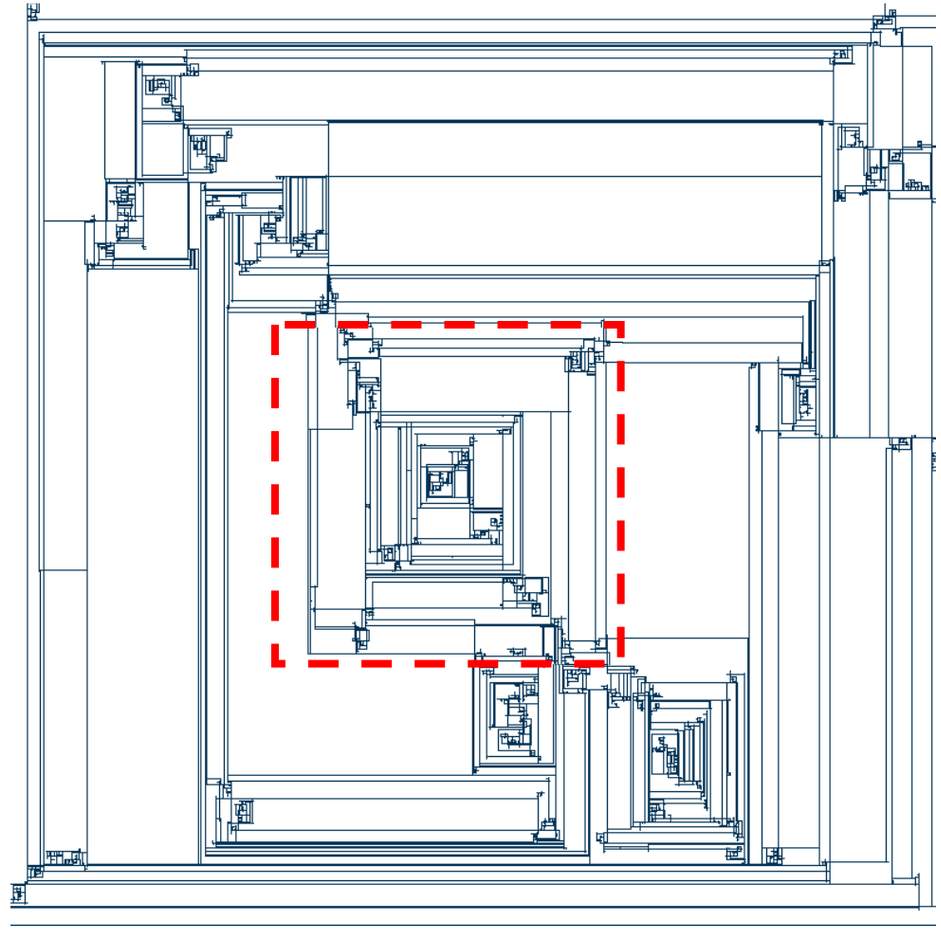
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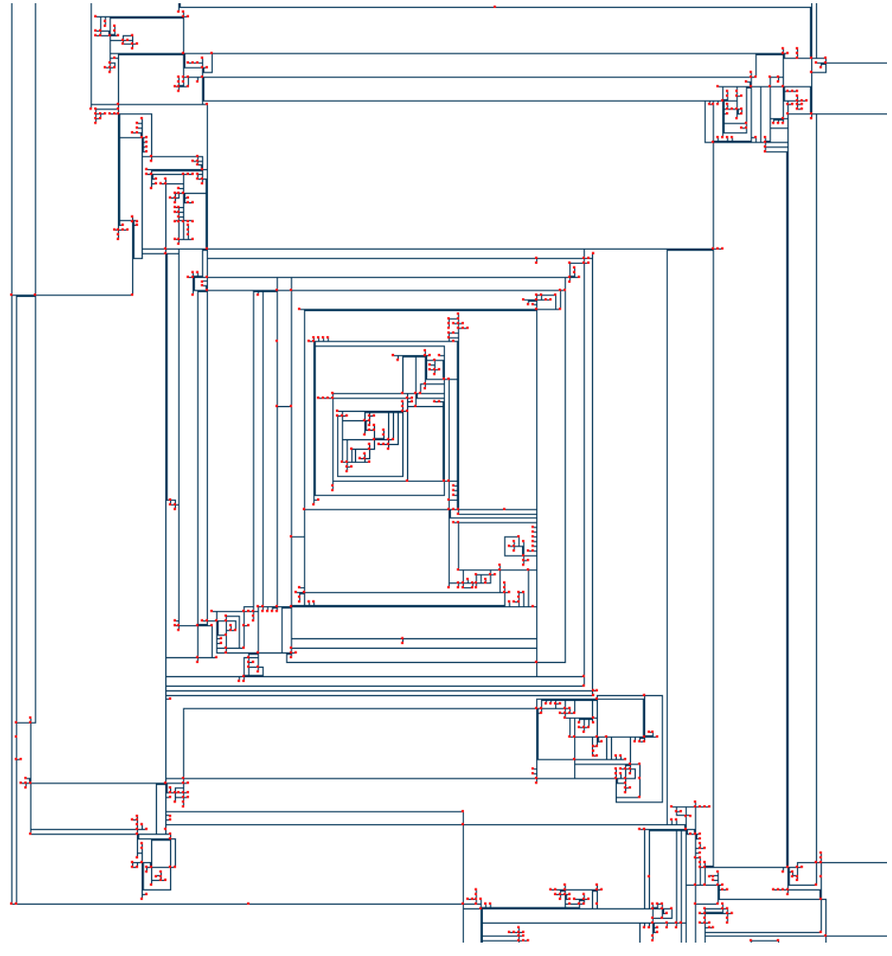
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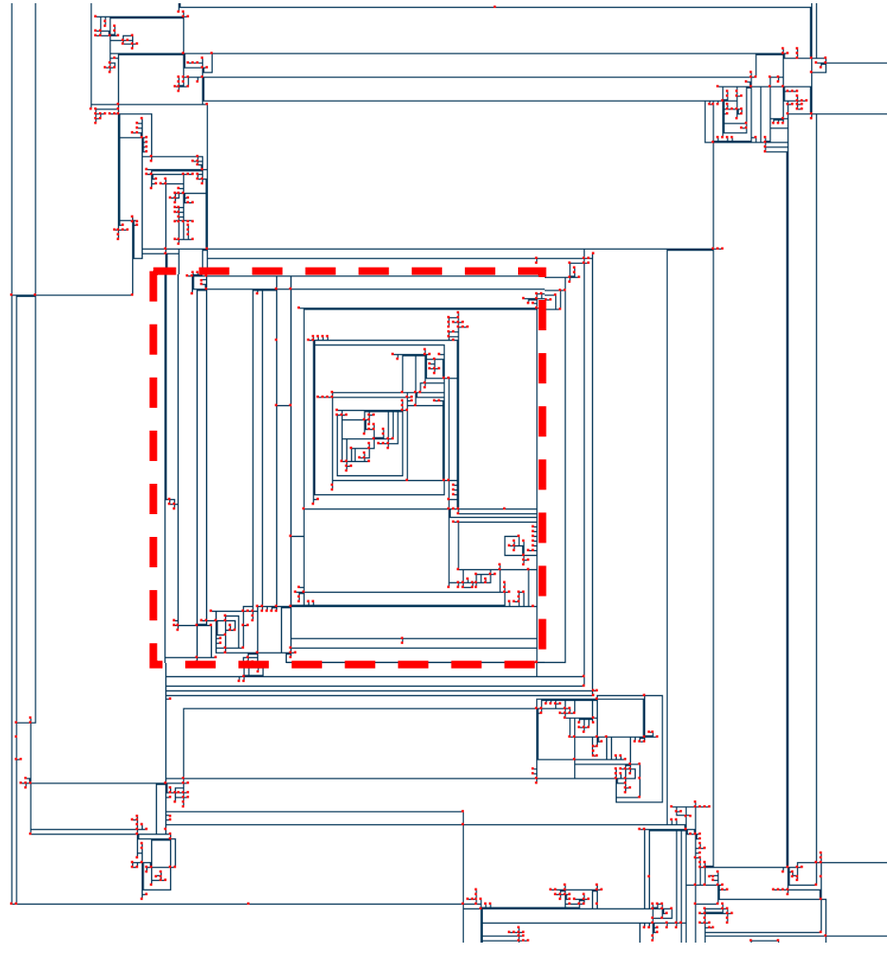
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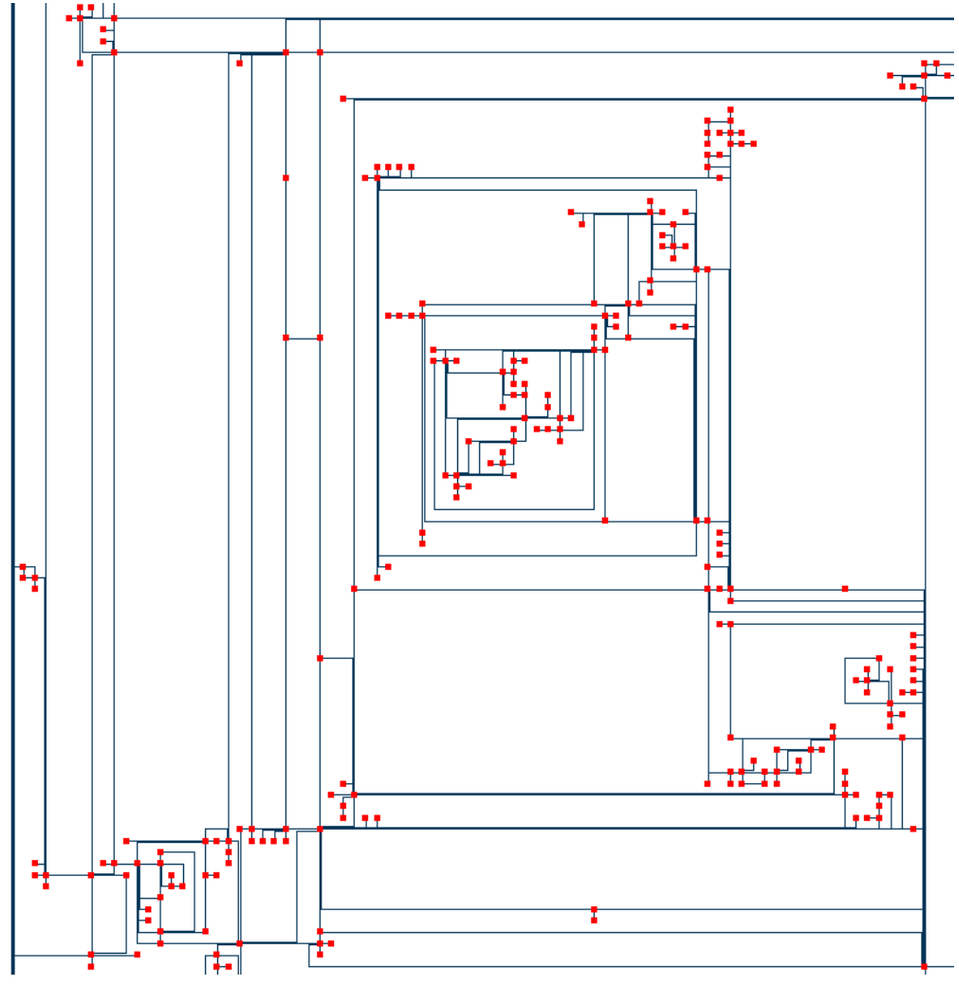
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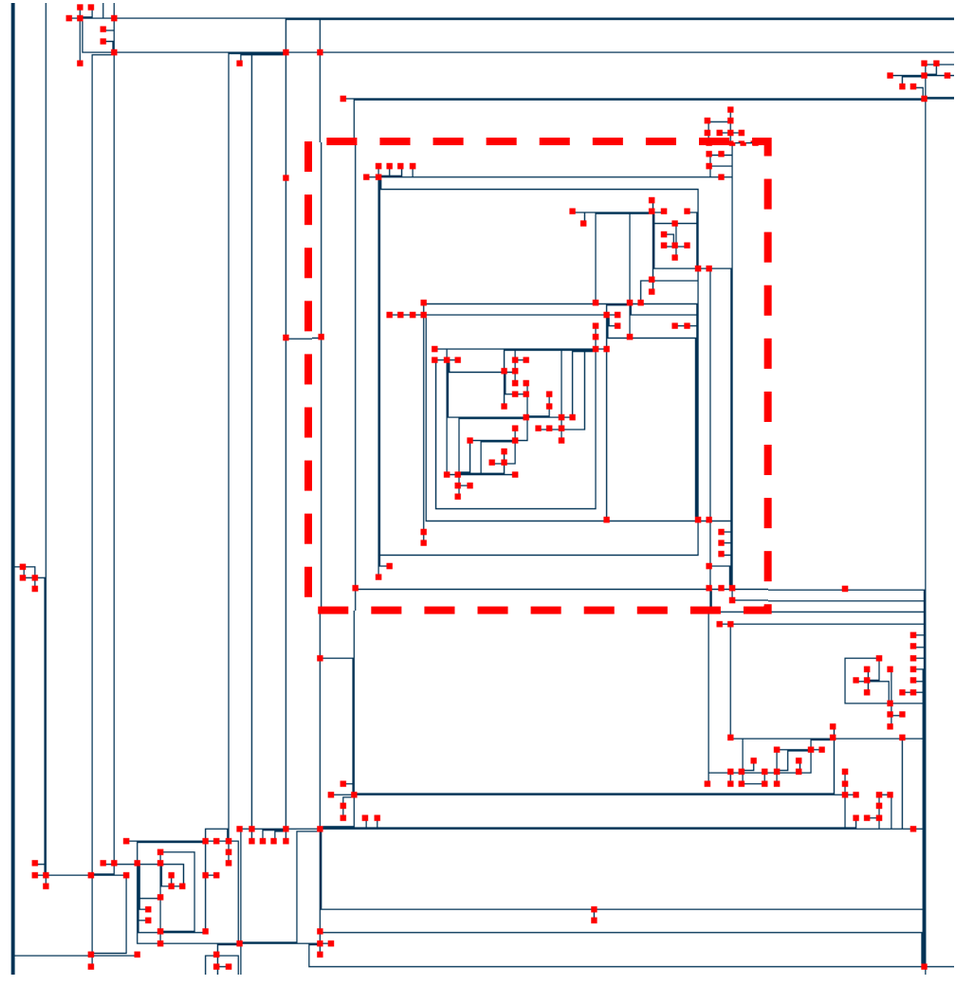
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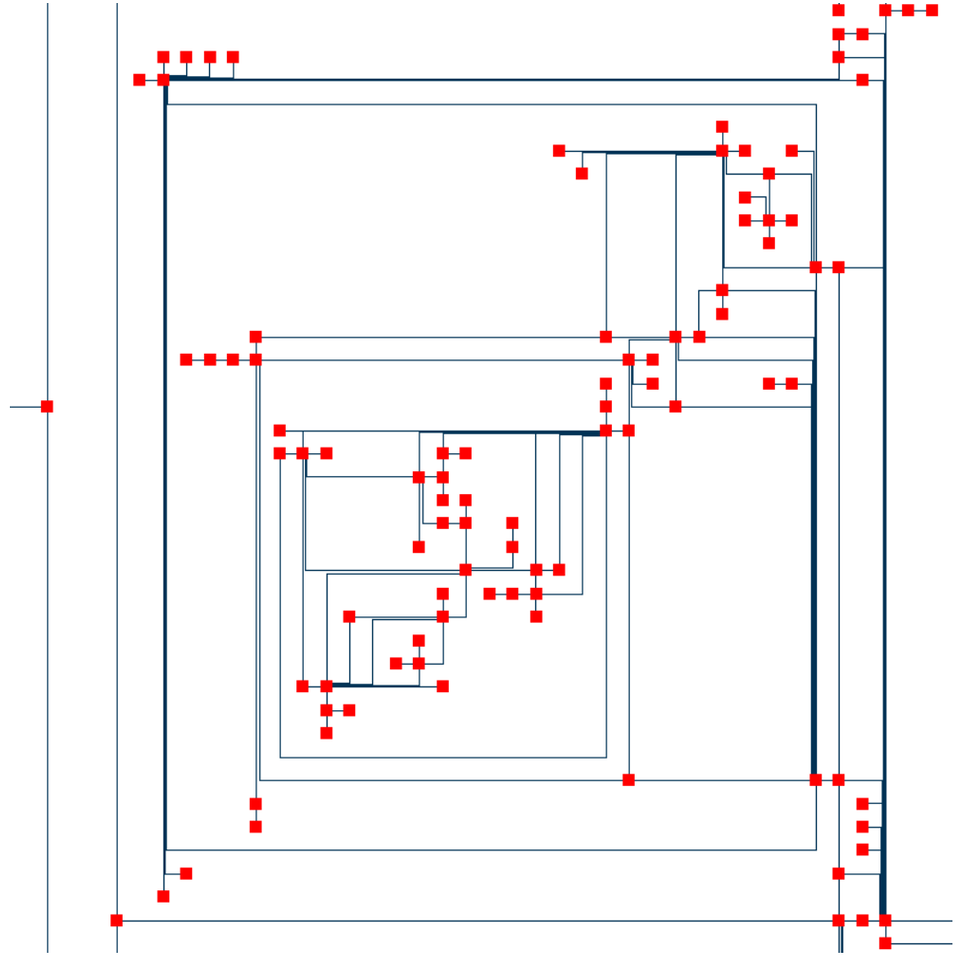
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Orthogonal drawings and self similarity

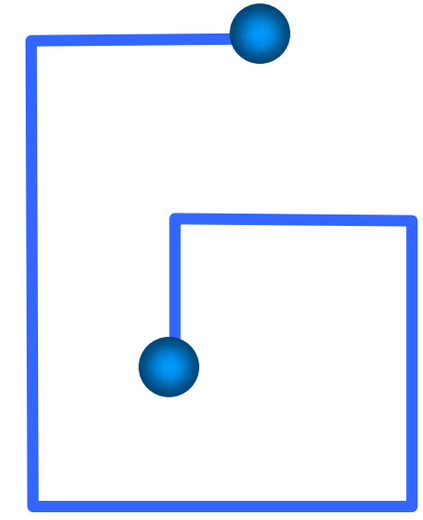


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Orthogonal drawings and bends per edge

- Tamassia's approach is based on a mapping with a flow network problem
- The flow network problem is solved by adding one unit of flow at a time
- The addition algorithm explores the graph and tends to use the same edges each times
- Bends are more likely to be pushed on the same edges



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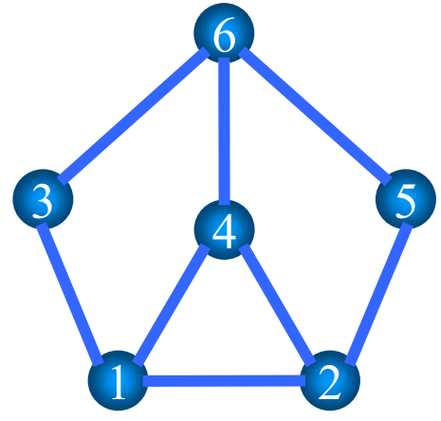
Alternative drawing algorithms

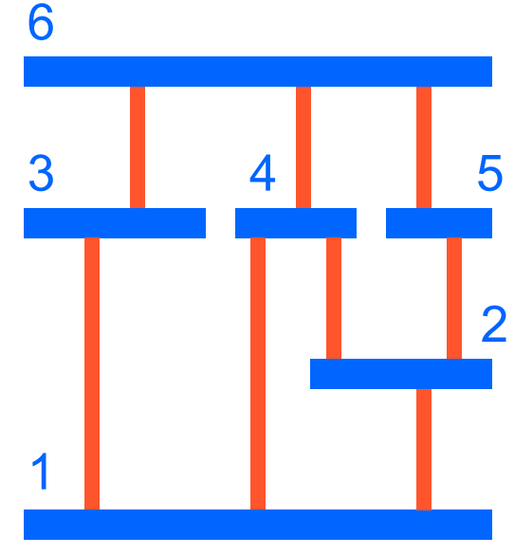
- In order to have a guarantee on the maximum number of bends per edge we consider
  - orthogonal drawings via visibility representations
  - incremental algorithms

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Visibility representation

- A *visibility representation* Γ of G maps every node v to a horizontal segment Γ(v) and every edge (u,v) to a vertical segment Γ(u,v) such that
  - each vertical edge segment Γ(u, v) has its endpoints lying on the horizontal vertex segments Γ(u) and Γ(v)
  - there is no other segment intersection or overlap

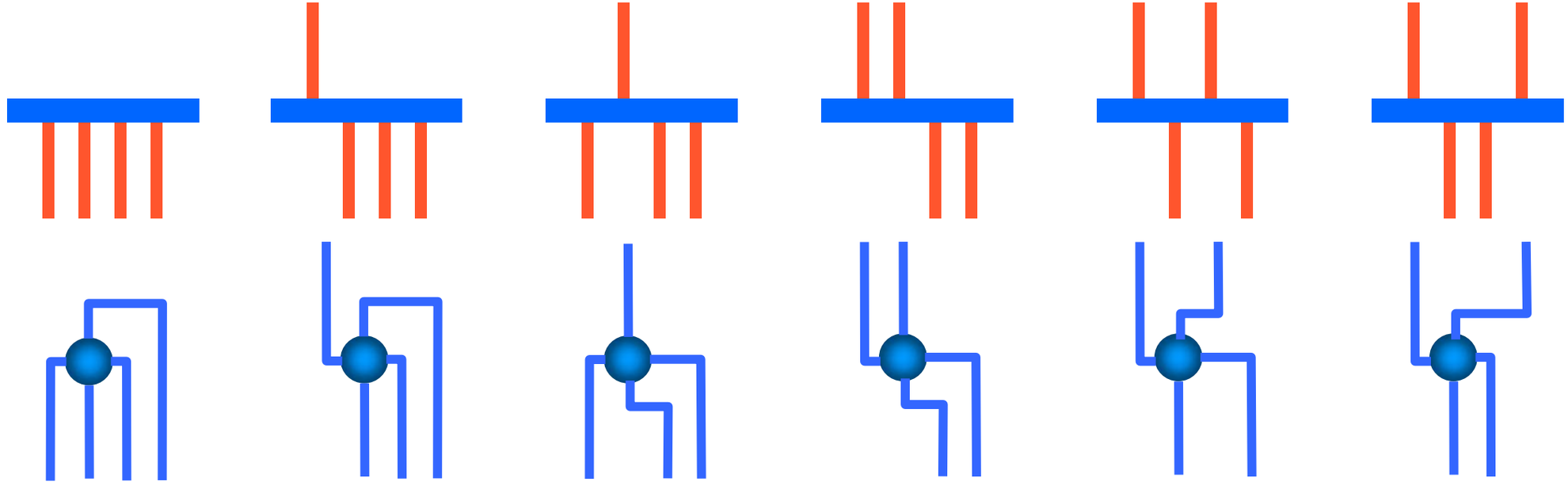




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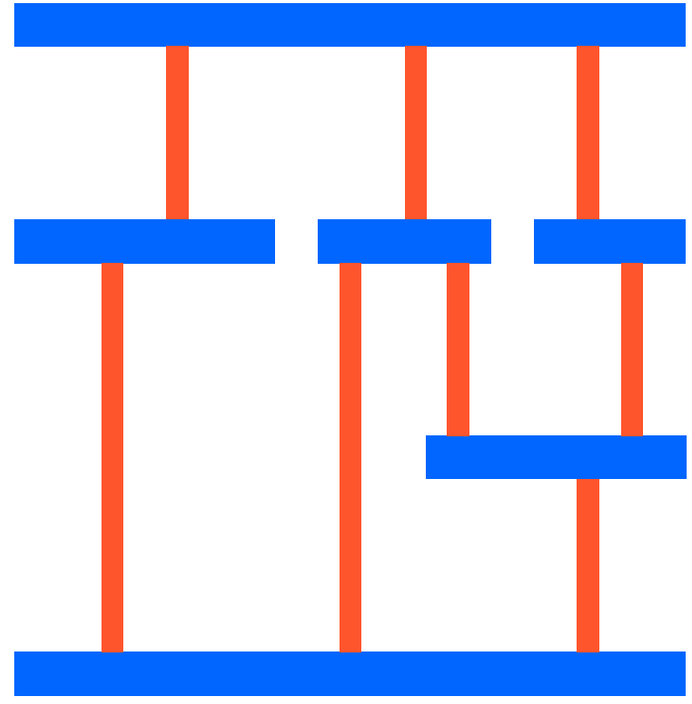
Segment replacement

- Segments can be locally replaced with nodes and edges in such a way to have an orthogonal drawing
  - example of replacement patterns



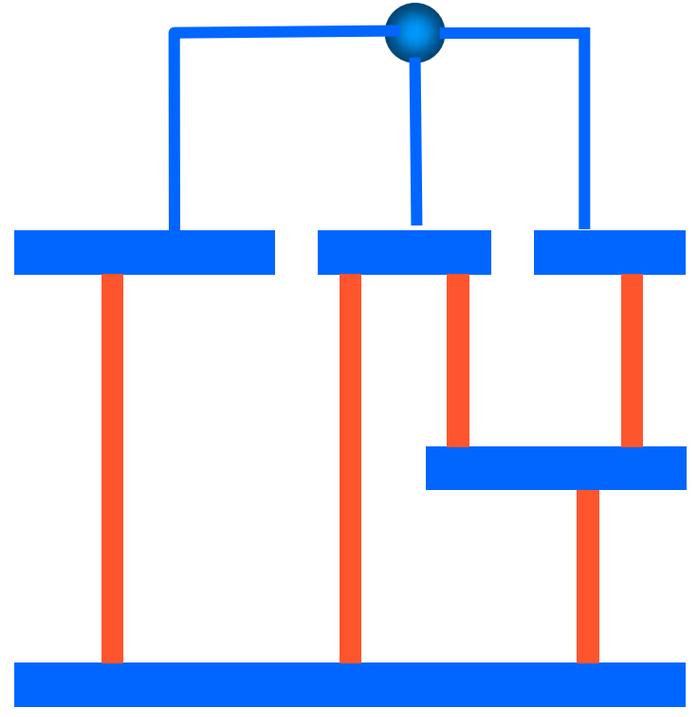
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Example



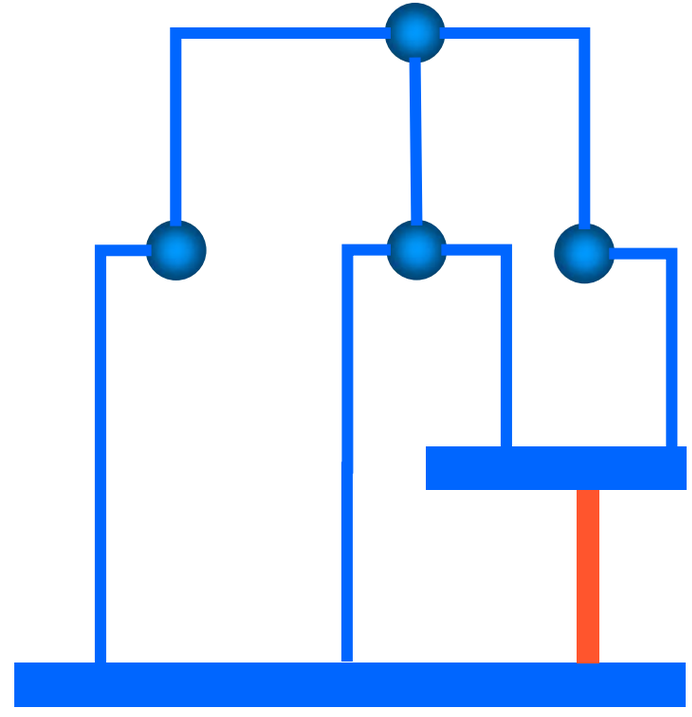
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Example



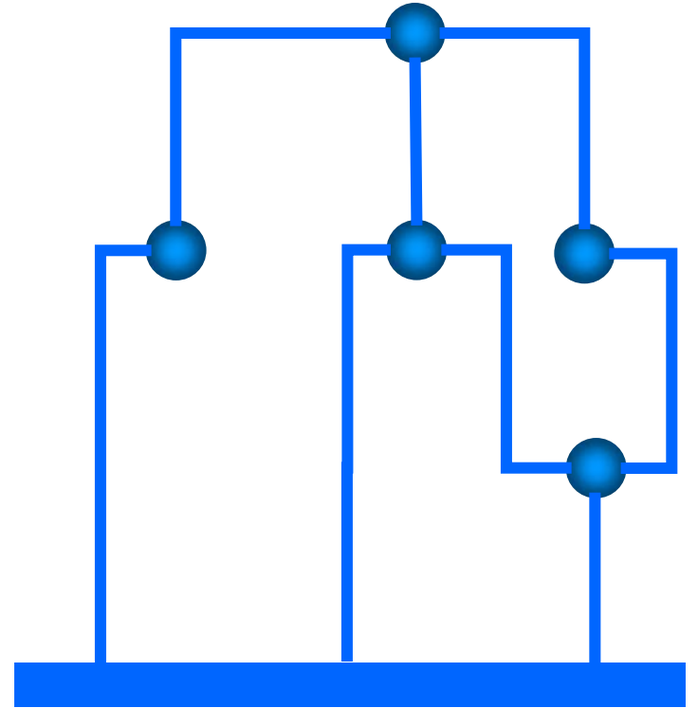
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Example



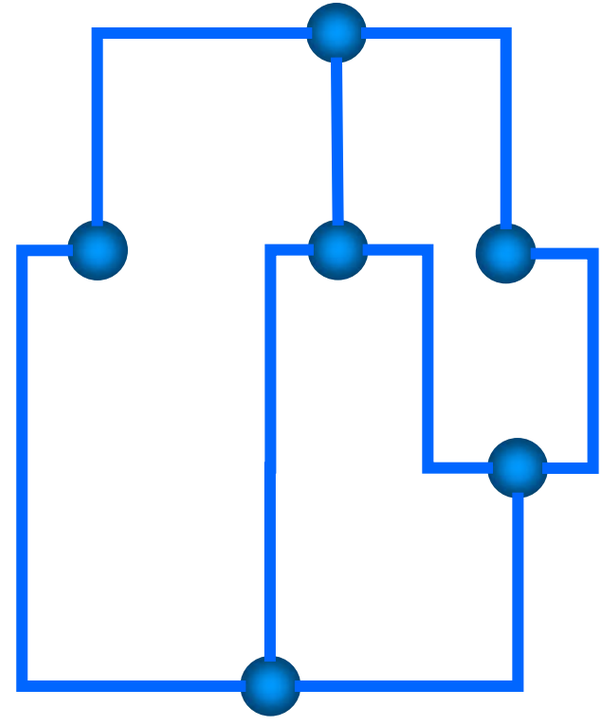
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Example



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Example



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Computing visibility representations

- Existence of visibility representations [Tamassia & Tollis, 86]
  - a graph admits a visibility representation if and only if it is planar
  - a visibility representation for a planar graph can be constructed in linear time

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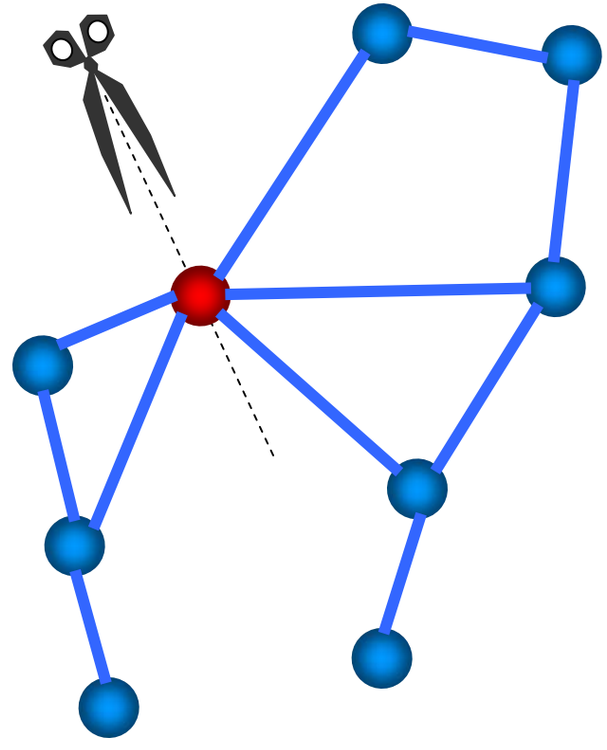
Computing visibility representations

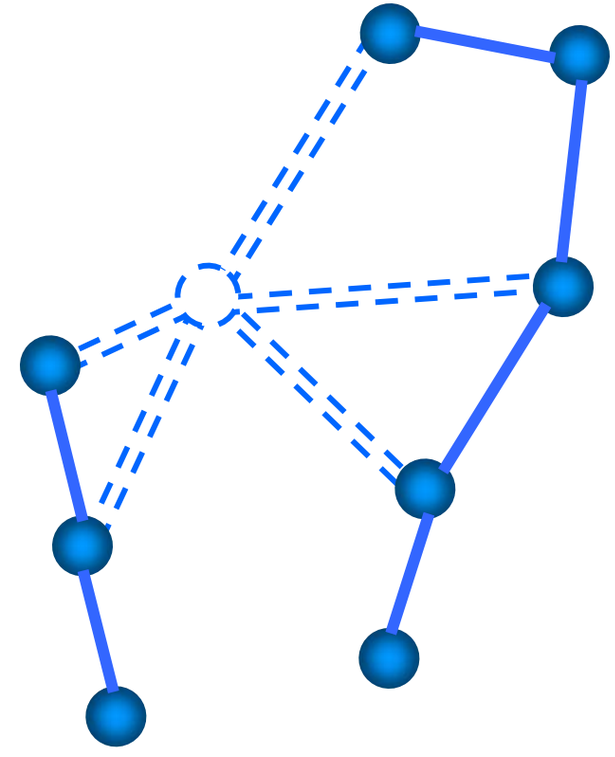
- Ingredients of the algorithm by Tamassia & Tollis for computing visibility representations
  - cut vertices and biconnected graphs
  - topological orderings
    - computing optimal topological orderings
  - planar st-graphs
  - dual of a plane graph
    - dual of a plane st-graph

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Recall: cut vertices

- A *cut-vertex* is a vertex such that its removal produces a disconnected graph
- A graph with no cut vertex is *biconnected*

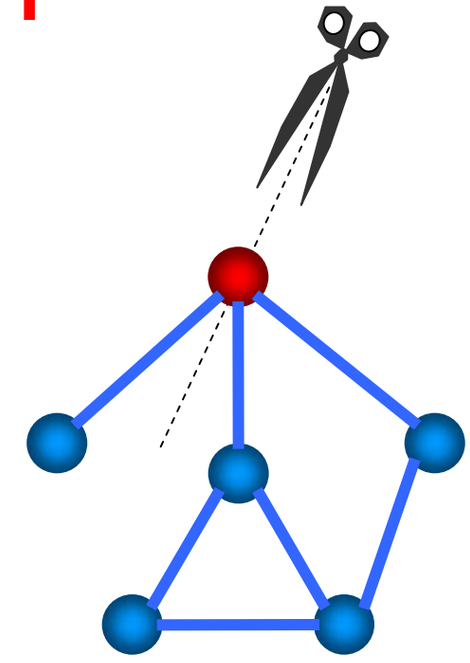


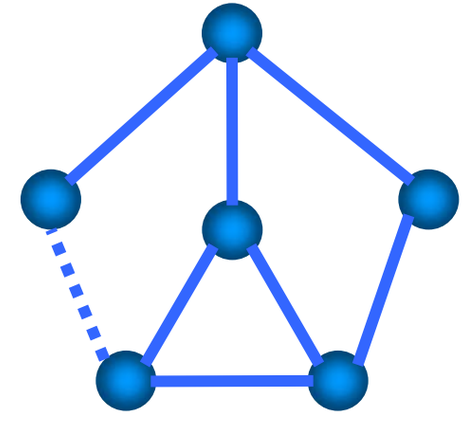


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Recall: biconnected graph

- A planar graph can be made biconnected (preserving planarity) by adding a linear number of dummy edges
  - [Fialko & Mutzel, 98]
- Hence, we can assume to always have a biconnected graph
  - add dummy edges
  - draw the graph
  - remove dummy edges

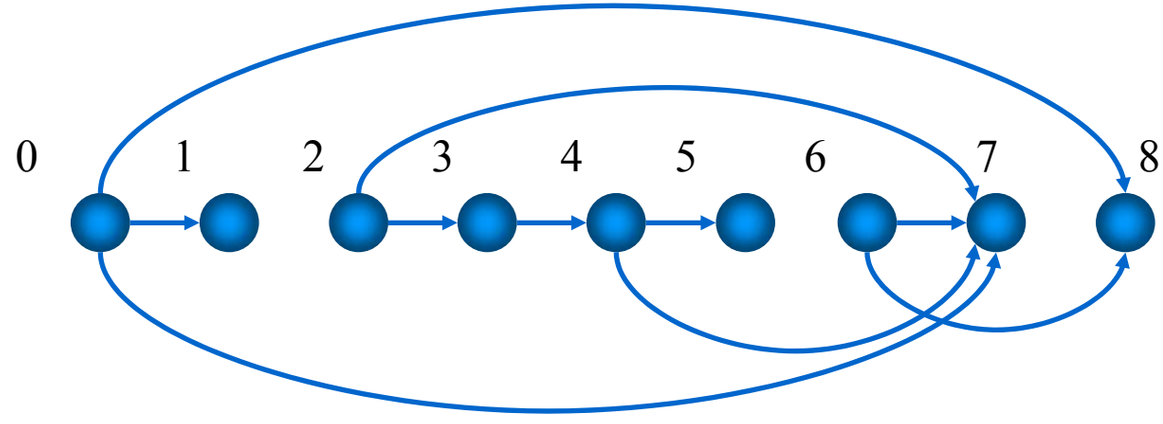




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Recall: topological sort

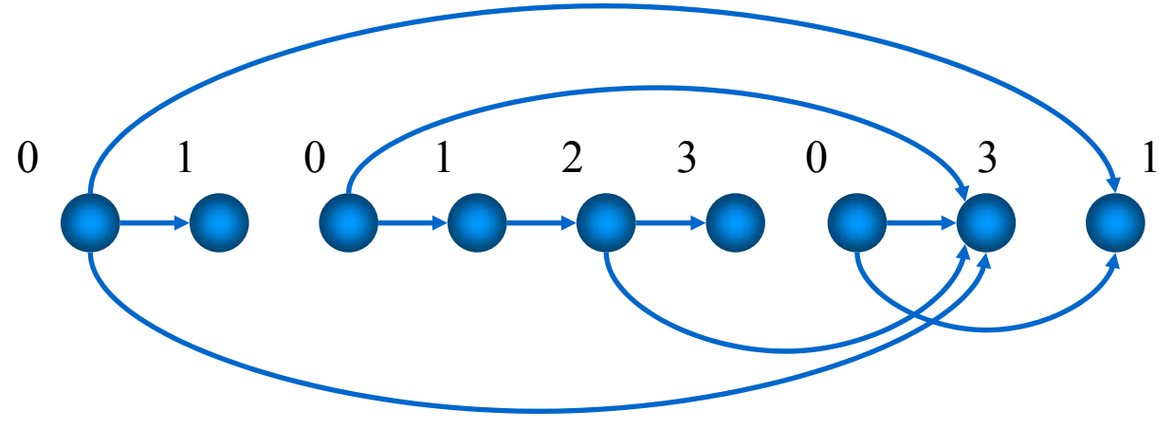
- Let G = (V, E) be a directed acyclic graph
- A *topological sort* of G is a linear ordering of its nodes such that for every directed edge (*u*,*v*), *u* comes before *v* in the ordering



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Recall: topological orderings

- Let G = (V, E) be a directed acyclic graph
- A *topological ordering* T(G) is an assignment of integer values T(v) to each node v such that for every directed edge (u, v) $\in$ E, we have that T(u) < T(v)



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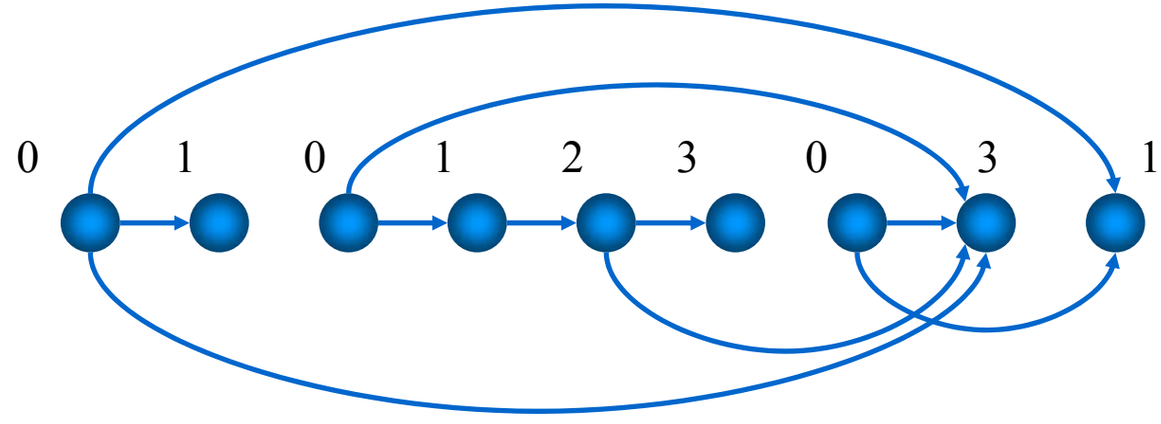
Recall: optimal topological ordering

- The *size* of the topological ordering is

$$\max_{v \in V}\left(T(v)\right) - \min_{u \in V}\left(T(u)\right)$$

- An *optimal topological ordering* T*(G) is a topological ordering with the smallest size

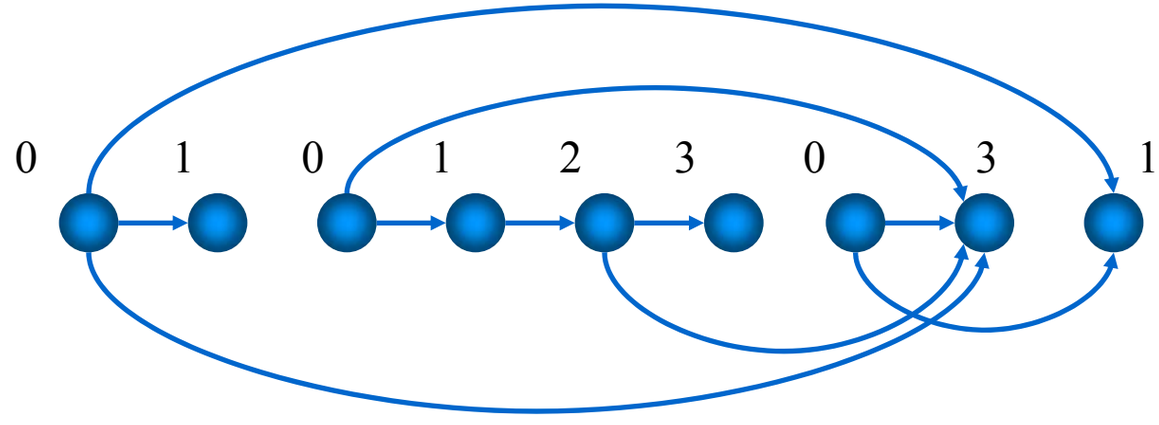
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Recall: computing topological orderings

- Computing an optimal topological ordering of a DAG in linear time is simple
  - assign every source node the number s=0
  - remove source nodes and their edges from the graph
  - repeat with the number s+1 until there is no node left



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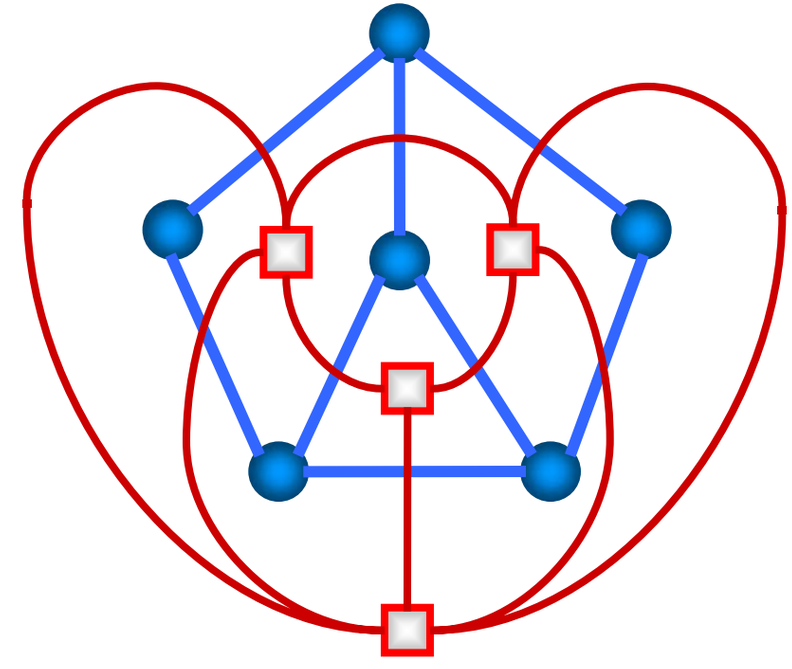
Planar st-graph

- An *st-graph* is a directed acyclic graph (DAG) with one source *s* and one sink *t*
- A *plane st-graph* has a planar drawing with nodes *s* and *t* on the external face
- Any biconnected graph admits an st-orientation with respect to any pair *s*, *t* on the external face
  - [Lempel *et al.*, 67]
- An st-orientation can be computed in linear time
  - [Even & Tarjan, 76]

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Dual of a plane graph

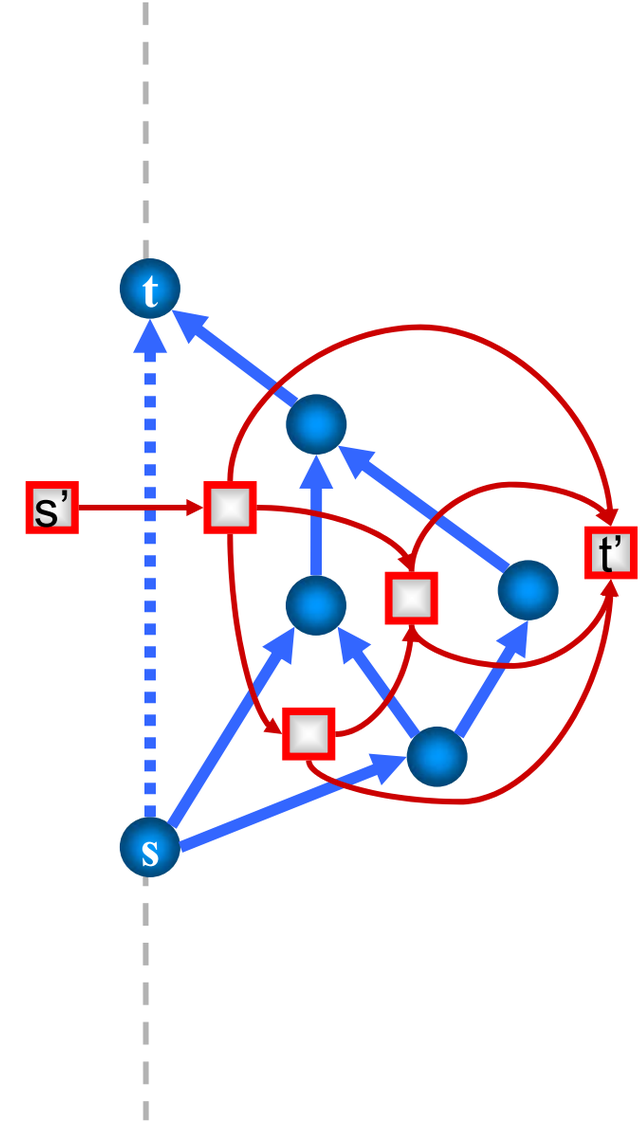
- Given a plane graph *G*, the *dual of G* is a (multi)-graph that has
  - one vertex v_f for each face *f* of the *primal* graph *G*
  - one edge between two vertices $v_{f'}$, $v_{f''}$ if the faces *f'* and *f''* are adjacent in *G*
- The dual of the dual is again the primal graph *G*



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Dual of a plane st-graph

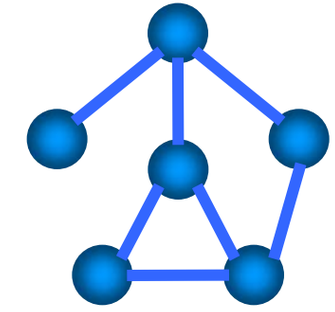
- Given a plane st-graph *G*, the *dual of G* is an st-(multi)-graph that has
  - one vertex v_f for each face *f* of the *primal* graph *G*
    - but it has two external vertices s' and t' as if the external face of G was divided into two faces
  - one edge between two vertices $v_{f'}$, $v_{f''}$ if the faces *f'* and *f''* are adjacent in *G*
    - the edges of the dual cross the edges of the primal from left to right

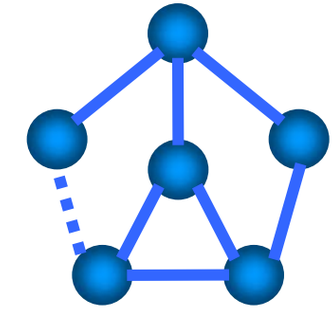


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Computing a visibility representation

1. Compute a planar embedding of the input graph
   - if needed, biconnect the graph by adding dummy edges
     - the degree of the nodes may become greater than four
     - after computing the visibility representation you remove the dummy edges and obtain again a maximum degree four graph

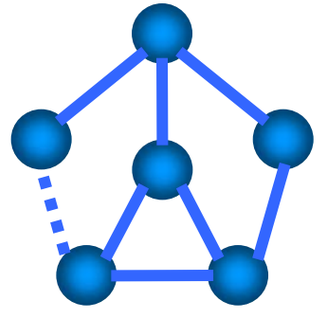


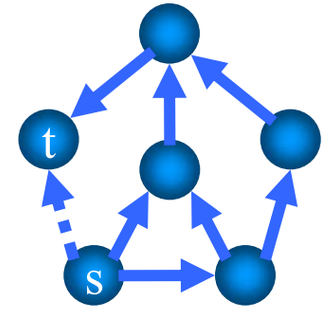


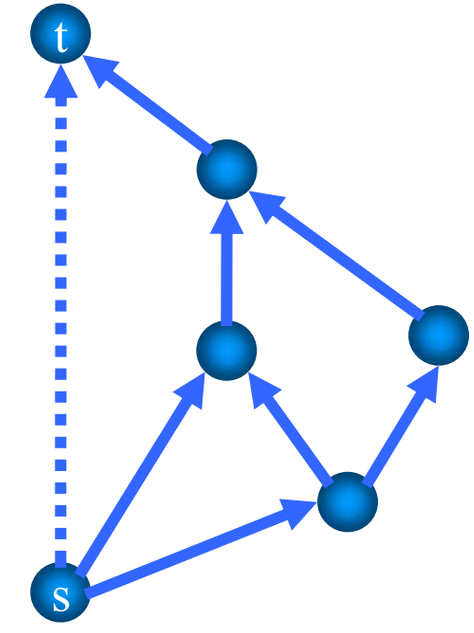
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Computing a visibility representation

2. Choose two nodes *s* and *t* on the external face and orient the edges to obtain a plane st-graph



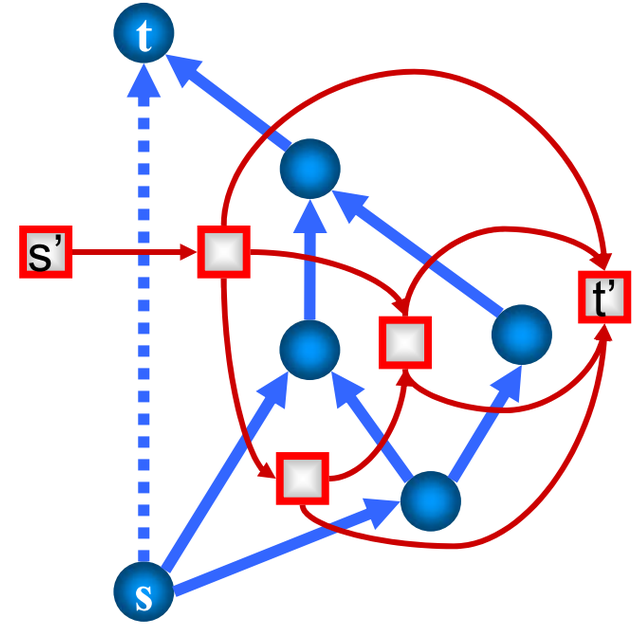




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Computing visibility representations

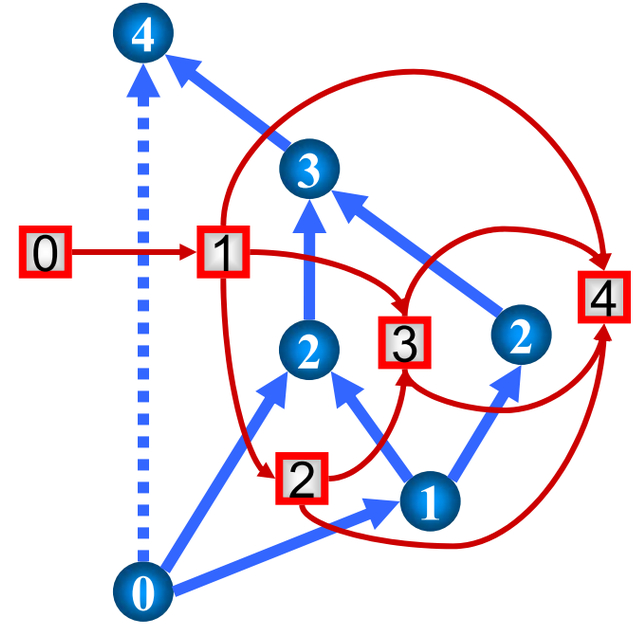
3. Compute the dual plane st-graph



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Computing visibility representations

4. Compute an optimal topological ordering for the graph and for the dual

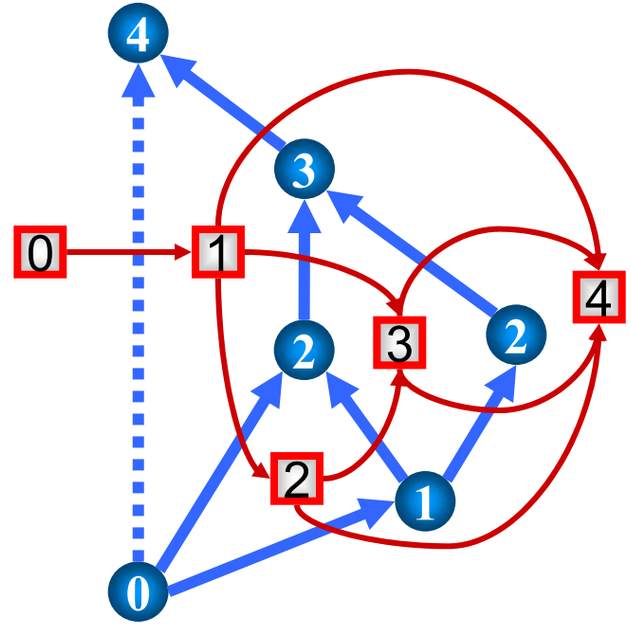


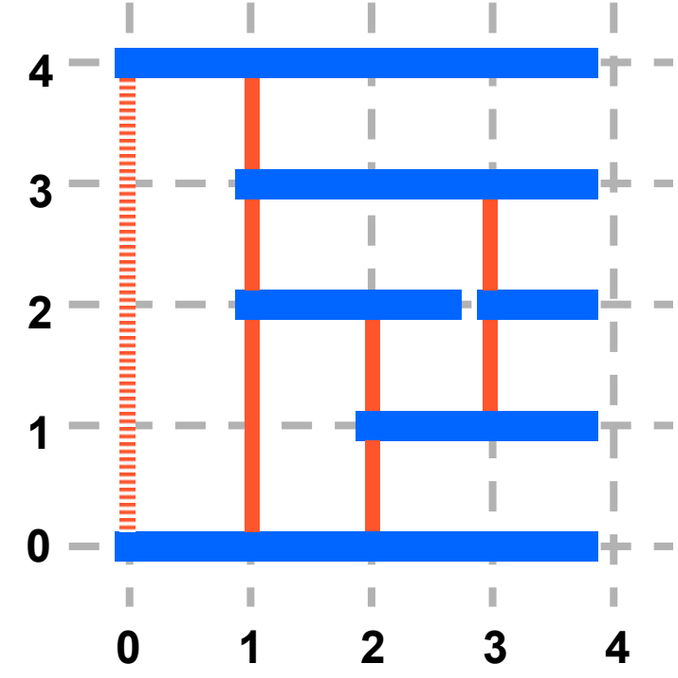
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Computing visibility representations

5. Construct a visibility representation

- can you guess the rule?

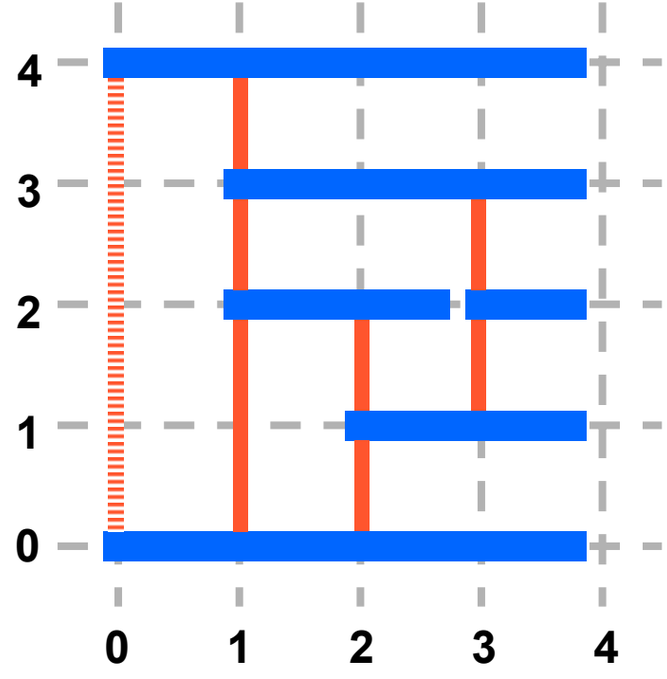


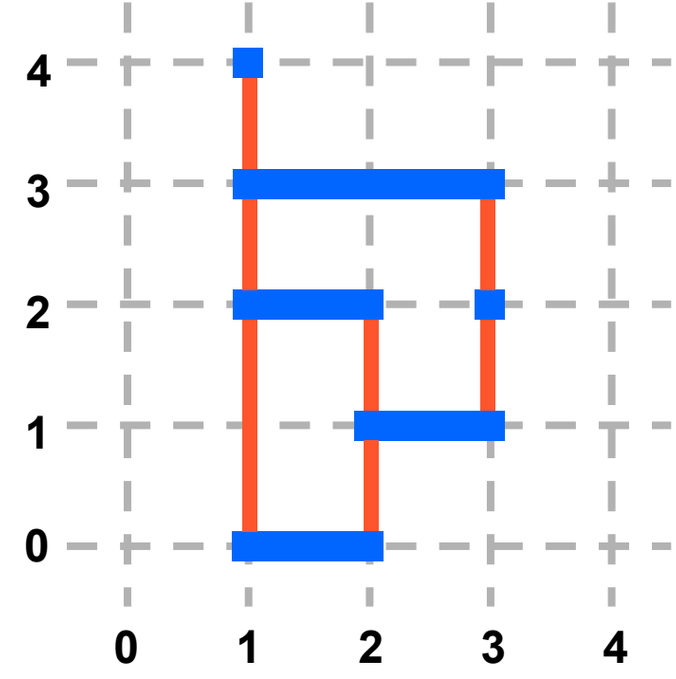


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Computing visibility representations

6. Remove dummy edges and simplify the representation

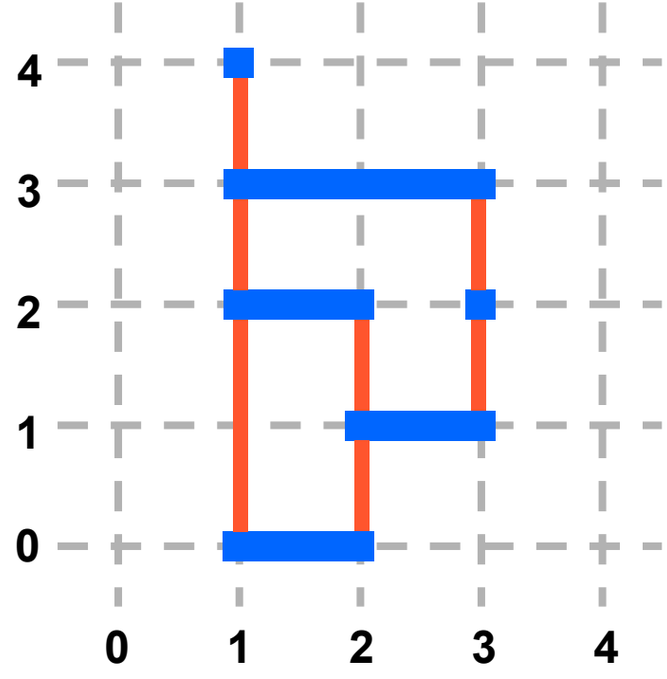


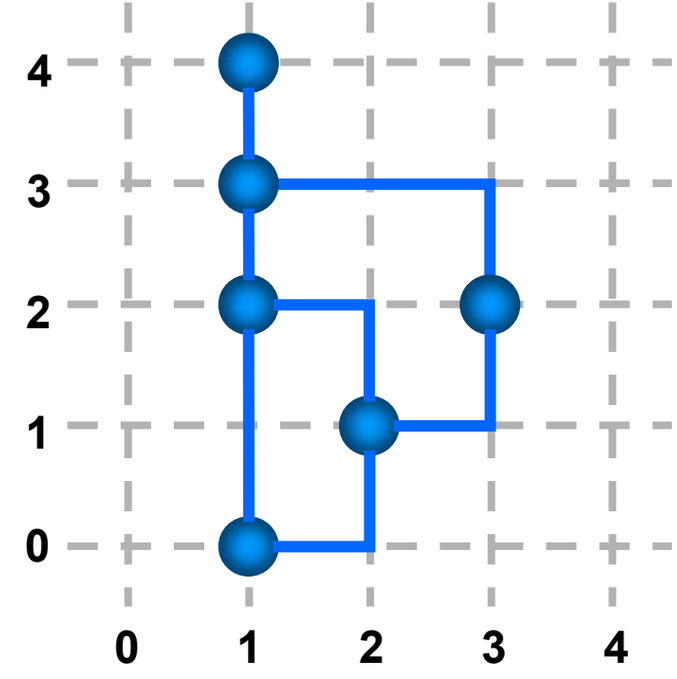


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Obtaining the final drawing

- Replace segments with nodes and edges to obtain an orthogonal drawing

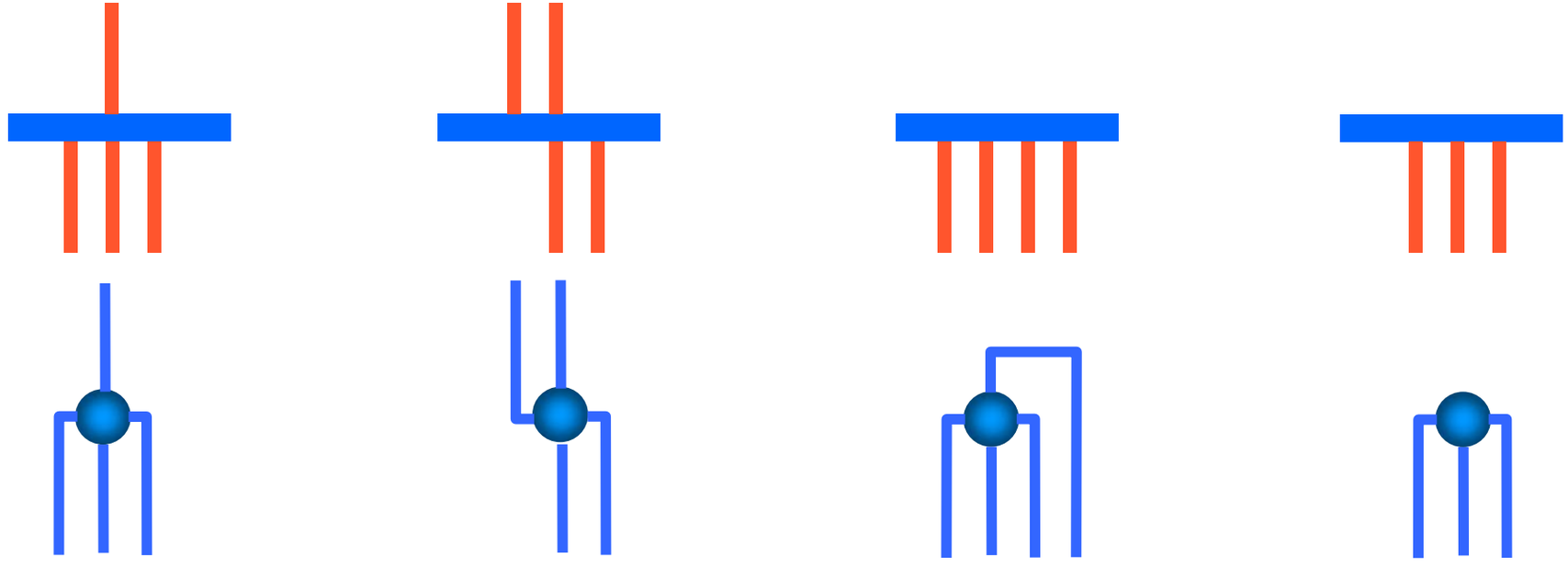




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A more sophisticated technique

- [Tamassia & Tollis, 89] propose a more sophisticated technique that only uses the following replacements
  - where the last two may only occur for *s* or *t*



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A more sophisticated technique

- The technique of [Tamassia & Tollis, 89] is based on computing a constrained visibility representation
- It can be proven that
  - if G is biconnected, there exists an orthogonal grid drawing of G using $O(n^2)$ area with at most 2n+4 bends and where only two edges have more than two bends
    - if G is connected, the number of bends is 2.4n+2 and no edge has more than four bends

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Bibliography and further readings

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